

Shyam J | AI/ML Engineer

Research-driven developer

Chennai, Tamil Nadu, India

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Professional Summary

AI/ML Engineer and Research-driven developer with hands-on experience across generative AI, digital twins, computer vision, and applied machine learning. Completed **5+ internships** and gained **1 year of industry experience at Sentient Scripts**. Proven ability to build ML pipelines, RAG systems, predictive maintenance models, and deploy full-stack AI solutions for real-world users.

Research Interests

Agentic AI, Reinforcement Learning, Generative AI, Digital Twins, Industry 4.0, Statistical Modelling, AI-powered Automation.

Technical Skills

Languages: Python, C, Java, TypeScript, JavaScript, SQL, Dart
Frontend: React.js, Next.js 14, Tailwind CSS, Bootstrap, HTML5, CSS3
Backend: Node.js, Flask, Express.js, REST APIs
AI/ML: PyTorch, TensorFlow, Keras, Hugging Face, Scikit-learn, RAPIDS cuML, FAISS, RAG systems, OpenCV
Analytics: Pandas, NumPy, Matplotlib, SPSS, Power BI, Hypothesis Testing
Cloud/DevOps: AWS, Oracle Cloud, Docker, Git, CI/CD, Serverless, Streamlit, Vercel
Databases: MongoDB, SQLite, Redis
Tools: VS Code, Postman, AutoCAD, Canva, Figma, BeautifulSoup, PyTesseract
Domains: Machine Learning, Deep Learning, NLP, Computer Vision, Digital Twins, Industry 4.0

Professional Experience

Linucare / Protechme

Embedded Firmware Engineer

Denmark (Remote)

May 2025 – Present

Working on embedded firmware for low-power connected devices

- Developing ESP32-based firmware modules, including device communication logic and peripheral management.
- Optimising system power consumption through efficient task scheduling and hardware-level configuration.
- Enhancing device performance by improving signal acquisition, data handling, and system responsiveness.
- Conducting power profiling across device states and contributing to core firmware documentation and architecture decisions.

Sentient Scripts Pvt. Ltd.

Student Coordinator / HR Lead

Remote

Jun 2024 – Jun 2025

Primary point-of-contact for SRM University partnership

- Led student recruitment, shortlisting, assessment coordination, and onboarding for multiple project cohorts.
- Managed documentation and communication for 100+ candidates.
- **Technical Work:** Developed a Generative-AI humorous weather chatbot with full-stack deployment.

Tata Steel

AI/ML Intern

Remote

Jan 2025 – Jun 2025

Guided by Dhilip Kumar R

- Implemented computer-vision pipelines for steel bar ring/rib analysis.
- Automated dimensional measurement using CNN-based feature extraction.

Chennai Metro Rail Limited (CMRL)

AI Intern

In collaboration with Mr. Ganavel, Cooling Lab

- Built predictive maintenance and fault-detection ML models.
- Delivered a troubleshooting tool.

Mercedes-Benz MBition

AI/ML Intern

Guided by Mr. Karthik VJ

- Contributed to digital-twin optimisation for NVIDIA Parker hardware.
- Supported AI-driven improvements to simulation performance.

Renault Nissan Automotive India Pvt. Ltd. (RNAIPL)

AI/ML Intern

Guided by Mr. Harishankar Girirajan

- Delivered supply-chain dashboards in Power BI and automated reporting workflows using UiPath and Power Automate.
- Developed analytics pipelines for Milkrun route optimisation, reducing routing delays and enhancing operational efficiency.

Sentient Scripts Pvt. Ltd.

AI Intern

- Built a Retrieval-Augmented-Generation (RAG) humorous Weather Chatbot achieving 89% user satisfaction.
- Developed a Streamlit dashboard for monitoring user interactions.

Education

SRM Institute of Science and Technology

B.Tech in Artificial Intelligence

- Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Data Structures and Algorithms, Database Systems, Software Engineering, Reinforcement Learning, Cloud Computing.

St. Michael Academy

Secondary & Higher Secondary Education

Certifications

Professional: Generative AI Professional (2024); Oracle Cloud Infrastructure GenAI (2024)

Online: AWS Machine Learning; Python for Data Science; AWS Cloud Foundations

Projects and Research

AI Educational Assistant

Edu Smart Bot

- Built OCR + transformer summarisation + context-based Q&A using BERT/T5.
- Developed Flask backend and React frontend; deployed on Vercel.

Autonomous Driving

Self-Driving Car AI Simulation

- Designed a Deep Q-Learning agent with CNN-based lane perception.

Vision-Language Model

Image Captioning System

- Built a CNN-LSTM attention model; achieved BLEU-4 score of 0.32.

Tamil Computing Centre (TCC), SRMIST

In-House Research

- Developed Tamil language models, CNN-based OCR, and AR/VR visualisation for Tamil literature.

Hybrid

Dec 2024 – Apr 2025

Berlin (Remote)

Nov 2024 – Apr 2025

Hybrid

Apr 2024 – Oct 2024

Remote

Apr 2024 – Oct 2024

Kattankulathur

Chennai

Feb 2025 – Jun 2025

Feb 2025 – Jun 2025

Feb 2025 – Jun 2025

Chennai

2025

Hackathons

- **Hackstreet 3.0 (2025)** — 1st Place for real-time AI automation.
- **Appathon 2.0 (2025)** — Winner (150+ teams).
- **Tamizh-A-THON 1.0 (2025)** — Winner for OCR NLP innovation.

Leadership Experience

NEXT GEN AI, SRMIST KTR

President

2025 – 2026

- Leading 400+ members; managing university-level industry collaborations.

NEXT GEN AI, SRMIST KTR

Vice President

2024 – 2025

- Led 350+ members; organised 5 major events.

AI Research Group

Technical Lead

2023 – 2024

- Mentored an 8-member research team in ML and CV.

CINTEL Students Association, SRMIST

Research Member

2023 – 2024

- Conducted applied AI research for departmental projects.

Additional Information

Languages: English (Fluent), Tamil (Native), Hindi (Conversational), Telugu (Fluent)

Professional Skills: Leadership, Team Management, Problem Solving, Critical Thinking, Communication, Time Management